Discover: Enzyme Action: Preparing the pH Investigation

Today I can… □ Use these words in the right way. Use them to explain an idea: enzyme, substrate, active site and specificity. □ Say how an unsuitable pH can change the shape of an active site. Say how this could slow the enzyme's work. □ Name the things in the test: pH is what we change. It is the independent variable. Time until starch disappears is what we measure. It is the dependent variable. Choose suitable control variables. These are things to keep the same.

Feedback: □ VF □ WS □ I □ NS □ E □ R □ [//] □ [?] _____

STAFF ANSWERS · Supported

Arrival

1. Nucleus.
2. From higher to lower concentration.
3. Water.
4. Active transport.

Exit

Starch fits the complementary active site, and amylase is not used up, so it can act again.

Change pH, time the first orange-brown sample in seconds, and match temperature or enzyme/starch amounts and concentrations.

Readiness bridge

It speeds up a reaction. It is not used up. We learn about enzyme action today. This fact is given to help us start.

Starch. Amylase breaks starch into smaller sugars.

Change only the pH. Keep the temperature the same. Keep the other relevant conditions the same too.

pH. Time is the measured outcome, or dependent variable.

Supported independent task

1. Amylase: enzyme; starch: substrate; not used up.
2. Change pH; measure endpoint time; keep temperature the same because it also affects activity.
3. F, B, D, A, C, E. F and B may swap if complete before mixing.
4. Endpoint: orange-brown remains. No-active-amylase comparison: blue-black, because starch should remain detectable.
5. Use the supplied four headings. Record "no endpoint by 300 s"; do not invent zero time or zero rate.

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STAFF ANSWERS · Independent

Arrival

1. Nucleus.
2. From higher to lower concentration.
3. Water.
4. Active transport.

Exit

Amylase's complementary active site binds starch, catalyses breakdown and releases products, leaving the enzyme available to act again.

IV is pH; DV is time in seconds to the first orange-brown sample; match temperature because it also affects enzyme activity.

Readiness bridge

It speeds up the reaction and is not used up, so it can act again.

Substrate: starch. Enzyme: amylase.

So a change in time or rate is not confused with a temperature effect. It makes the comparison more valid for investigating pH.

Independent: pH. Dependent: time to the defined reaction endpoint, measured in seconds.

Independent independent task

1. pH may affect activity, giving different endpoint times; the fastest tested condition is to be found. Products leave and amylase is not used up.

2. IV pH; DV time / s to the first sampled drop leaving iodine orange-brown. Controls include temperature, concentrations, volumes and sampling conditions. Temperature also affects activity.

3. F, B, D, A, C, E (F/B may swap). Separate iodine drops avoid changing the main reaction mixture with the detector.

4. A control variable is kept the same; a control sample is an additional comparison. Without active amylase starch should remain detectable, giving blue-black.

5. Record "no endpoint by 300 s"; leave numerical time/rate uncalculated for that run. 1/time compares matched starting amounts and endpoints; it is not product mass per second.

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STAFF ANSWERS · Aspire

Arrival

1. Nucleus.
2. From higher to lower concentration.
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Exit

Complementary binding enables catalysis, product release and enzyme reuse; unsuitable pH may alter active-site shape or catalysis, reducing activity without proving permanent denaturation.

Change mixture pH, time the first orange-brown sampled drop in seconds, and match temperature to exclude temperature as another cause of timing differences.

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STAFF ANSWERS · Aspire · continued

Readiness bridge

Amylase catalyses the reaction. It can act again. Starch is the substrate. It is changed into smaller sugar products.

Both factors could change the rate. A change in rate might not be due just to pH. Keep the amylase concentration the same.

Repeats show variation. They can help us spot a doubtful run. They help us give a more reliable summary of the time at that pH. Repeats alone do not remove a systematic error.

Different rules for when to stop can make the times look different. Use the same endpoint that you can observe. Then the comparison is more consistent.

Aspire independent task

1. pH may alter binding/catalysis, changing endpoint time. A complementary active site binds starch; products leave and the catalyst is not used up. Slower activity alone cannot establish permanent denaturation.

2. IV mixture pH; DV time / s to first orange-brown sample. Match temperature, enzyme/starch concentration and volume, total volume, start/mixing and sampling conditions. Link at least two to plausible rate/timing confounds.

3. F, B, D, A, C, E, allowing F/B swap. Separate iodine avoids altering the reaction; a consistent mixing/timer start prevents an artificial time difference.

4. A variable is matched; the control sample lacks active enzyme. Blue-black expected. If orange-brown, check starting starch, reagent function, sample identity and contamination before interpreting enzyme action.

5. Record 120 s; threshold crossed after 90 and by 120 s, not exactly 105 s. A shorter feasible fixed sampling interval reduces interval uncertainty. Record "no endpoint by 300 s"; no invented numerical rate.

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STAFF ANSWERS · shared tasks and mark scheme

Shared activities

Model: 1 amylase enzyme; 2 starch substrate; 3 active site; 4 smaller sugar products. The enzyme is not used up. Method order F, B, D, A, C, E, allowing F/B swap. Blue-black detects starch; the first orange-brown remaining sample gives the endpoint. No-active-amylase control expected blue-black.

Independent

1. Write a testable prediction for how pH could affect amylase acting on starch. Explain why the enzyme can act again. [2 marks]

Changing pH may change endpoint time under matched conditions (1 mark)

Amylase is not used up and can catalyse another reaction (1 mark)

Independent

2. Name the IV and define the timed DV with units. Give three controls and justify one. [6 marks]

pH is deliberately changed (1 mark)

Time to first orange-brown iodine sample is measured in seconds (1 mark)

Keep temperature constant (1 mark)

Keep enzyme concentration or volume constant (1 mark)

Keep starch concentration or volume constant (1 mark)

A named control could otherwise alter rate, so matching it isolates pH (1 mark)

Independent

3. Order the six method cards and explain why iodine is used to test separate sampled drops. [2 marks]

F/B before D, then A, C, E, with preparation before mixing (1 mark)

Sampling checks starch without adding iodine to the main reaction (1 mark)

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STAFF ANSWERS · shared tasks and mark scheme · continued

Independent

4. Distinguish a no-active-amylase control sample from a control variable. Predict the control result. [3 marks]

A control variable is a factor kept constant (1 mark)

The no-active-amylase sample is a separate comparison (1 mark)

It should stay starch-positive and turn iodine blue-black (1 mark)

Independent

5. State suitable table headings for pH, trial, endpoint time / s, observation and relative rate 1/time / s^{-1}. State what to record if there is no endpoint by 300 s. [3 marks]

Headings include pH, trial, endpoint time / s, observation and relative rate / s^{-1} (1 mark)

Record no endpoint by 300 s (1 mark)

Do not calculate an exact reciprocal rate for an unobserved endpoint (1 mark)

Aspire

1. Predict a pH–activity pattern that we can test. Explain specificity. Explain how the enzyme acts again. Do not claim that slower activity proves permanent denaturation. [4 marks]

A testable prediction relates changed pH to endpoint time (1 mark)

Substrate fits a complementary active site (1 mark)

Enzyme is not used up (1 mark)

Slower activity alone does not establish permanent denaturation (1 mark)

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STAFF ANSWERS · shared tasks and mark scheme · continued

Aspire

2. State the IV. Define a DV that someone else could measure in the same way. Give four controls. Justify two of them. Say what other cause each rules out. [8 marks]

pH is the independent variable (1 mark)

The measured variable is the first starch-negative sampled time in seconds (1 mark)

Match temperature (1 mark)

Match amylase concentration and volume (1 mark)

Match starch concentration and volume (1 mark)

Match total volume, starting conditions or sampling interval (1 mark)

Justify one control as excluding an alternative rate effect (1 mark)

Justify a second control as excluding an alternative rate effect (1 mark)

Aspire

3. Order method cards A–F. Explain why iodine goes in separate check drops. Explain why we must start in the same way. [3 marks]

F/B before D, then A, C, E (1 mark)

Iodine is kept out of the reacting mixture (1 mark)

A consistent mixing/start event keeps the timing comparison valid (1 mark)

Aspire

4. Show how a no-active-amylase control differs from a control variable. If its result is orange-brown when you did not expect that, what would you check? [3 marks]

Matched conditions are controlled variables (1 mark)

The no-active-amylase control is a separate comparison (1 mark)

Orange-brown would prompt a check for absent starch, contamination or an inactive detector before accepting results (1 mark)

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STAFF ANSWERS · shared tasks and mark scheme · continued

Aspire

5. Use these example samples: blue-black at 90 s and orange-brown at 120 s. Samples were 30 s apart. State the time to record. State the time window when starch crossed the detection threshold. Give one way to make timing more precise. A further run has no endpoint by 300 s. Choose a truthful table entry. [4 marks]

Record 120 s (1 mark)

Crossing occurred after 90 and by 120 s (1 mark)

A shorter authorised sampling interval narrows the timing window (1 mark)

Record no endpoint by 300 s for the other run, without an invented exact rate (1 mark)

Aspire

Accept scientifically equivalent wording. Award each distinct point once. The mark tariff is for the question, not a participation score. Keep the answer scheme staff-only during independent work.

Choice

Use the matching independent-task answers; preserve the pupil's own scientific sentence, label or reason. Staff mark I, S or C beside this task.